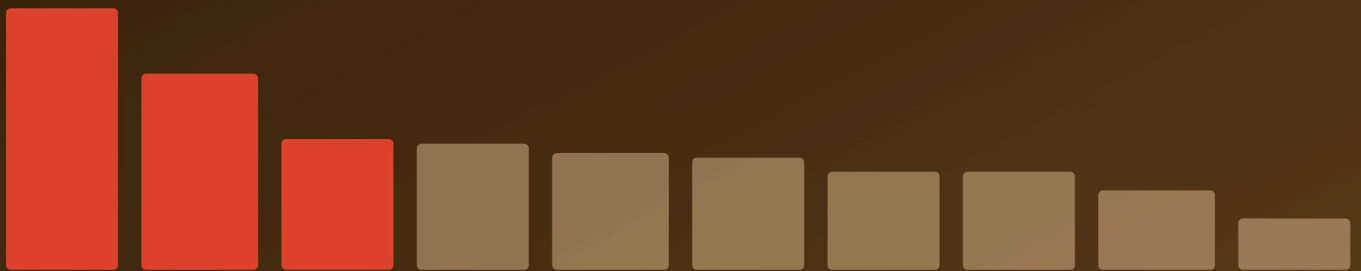


DEEP PYQ ANALYSIS

GATE ME 2027

High-Probability Question Bank
& 10 Full-Length Mock Tests



Marks per section · six counted papers

MD ANISH AHAMAD

GATE ME 2027 — Deep PYQ Analysis & High-Probability Question Bank

By MD ANISH AHAMAD · First Edition 2026

This preview contains 34 real, unedited pages from the 688-page book, exactly as they appear in the full PDF. Page numbers in the running heads and on the contents pages refer to the full book.

In this preview	In the full book	Pages
Title page	front matter	1
Contents	front matter	3
Preface	front matter	2
Executive summary	pages 1–3	3
What changed in the 2027 syllabus	pages 8–9	2
The organising institute	pages 16–17	2
How the six papers were counted	pages 20–21	2
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Sample questions with four confidence figures	pages 217–219	3
The hundred questions to work first	pages 314–315	2
What not to skip, and what to skip	pages 320–321	2
How to take and score a mock test	pages 326–327	2
Mock Test 01 (first pages)	pages 329–330	2
Mock Test 01 answer key	page 341	1
Mock test worked solutions (first pages)	pages 487–488	2
Last-minute sheet (first pages)	pages 647–648	2
How to check any number in this book	page 671	1

The full book also has: all eleven subject chapters, 292 practice questions with worked solutions and four confidence figures each, all ten mock tests with keys and solutions, the hundred to work first, the what-not-to-skip chapter, the revision plan and the full last-minute sheet.

GATE ME 2027

Deep PYQ Analysis & High-Probability Question Bank

Six GATE Mechanical Engineering papers counted question by question, a question bank in every subject chapter, and ten full mock tests built on what the papers show for the GATE Mechanical Engineering paper

MD ANISH AHAMAD

First Edition · 2026

For the examination of February 2027

gateguide.co.in

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Preface

Every GATE book tells you what is in the syllabus. The syllabus is three pages long and the organisers publish it free. This book tries to answer the question the syllabus cannot: ****of the eighty-five technical marks in the GATE Mechanical Engineering paper, where do they actually land — and how much of that can anyone honestly claim to know?***

The organisers have never published a marks breakdown. Every "subject-wise weightage" figure for ME is therefore somebody's count of some set of past papers, and a count is only as good as two things it rarely prints: which papers it covered, and where it put the questions that sit between two subjects. In a paper whose subjects share as much as Mechanical Engineering's do — a band brake that is both friction and a design element, a Coriolis component that is both rigid-body kinematics and mechanism analysis, a nozzle that is both fluid mechanics and thermodynamics — the second omission is not a small one.

So this book does not use anyone's table. It counts the papers. Every one of the last five years.

How this book was built

Six papers, counted question by question. GATE Mechanical Engineering was set as two papers in 2022 (Set 1 and Set 2) and as a single paper in each of 2023, 2024, 2025 and 2026. All six — 390 questions and 600 marks — were read one question at a time. Each question was assigned to one of this book's ten technical subjects, built on the paragraph labels of the official 2027 syllabus, and to a topic inside it, and its marks and question type were taken from the official answer key for its paper, never from the paper's layout. Each paper was made to close: 65 questions, 100 marks, 30 at one mark and 35 at two, General Aptitude at exactly 15. A count that does not close is a count with a mistake in it, and all six close.

Every citation names a paper and a question number. Not once in this book will you find a claim like "this pattern keeps recurring". A citation such as "2025 (Q35)" names exactly one question, because 2025 was one paper; a 2022 citation always names its set — "2022 S1 (Q35)" or "2022 S2 (Q35)" — because 2022 was two. Every historical statement in this book can be checked against a paper you can download from an organising institute's own website.

Every boundary decision is written down. 52 of the 330 technical questions could honestly have been counted in a second subject. Each records the subject it was counted in, the subject it could have gone to, and why, so that a reader who disagrees can move it and see exactly what changes. Thirteen were not settled by any stated rule, and each of those is marked as counted at its lean rather than presented as certain.

****The 2027 syllabus was compared with 2026, phrase by phrase and word by word, from the two official PDFs, by a program rather than by reading. The result is not small: 30 phrases added, 5 removed and 19 reworded**, two new paragraph labels — Additive Manufacturing, and Computer Aided Manufacturing and Automation — and two old ones, Computer Integrated Manufacturing and Inventory Control, folded into their neighbours. The same program compared the 2024, 2025 and 2026 documents and found them identical in wording. Section 2 prints the comparison and the method. A change list that is wrong is worse than none, so this one was generated from the source rather than copied from anyone's summary of it.**

Every number in every question was computed, not remembered. Beam reactions and bending moments were evaluated, cycle efficiencies were worked from the property values the question prints, fin and exchanger answers were solved from their equations, and every numerical-methods question was iterated exactly as its stem prescribes. Every incorrect option is the output of a specific mistake, and the solution names that mistake, so that a candidate who chose it learns what they did rather than merely that they were wrong.

What this book refuses to do

It does not claim to know the paper. Four commitments follow:

- **Facts, analysis and predictions are labelled separately.** A statement about what happened in 2025 is marked FACT. A statement about what the data suggests is marked ANALYSIS. A statement about 2027 is marked PREDICTION. A statement that is the author's opinion is marked JUDGEMENT. They are never blended.
- **Confidence is reported in four dimensions, not one.** Being confident that the second law will be examined is not the same as being confident about a particular entropy-generation question. Every question carries Topic, Concept, Question-pattern and Exact-question confidence, and the last is capped at 35 per cent because GATE repeats concepts, not wording.
- **The confidence figures are called estimates, because that is what they are.** They are not a fitted model and they have not been validated. Six papers could not validate them if anyone tried. Section 7 explains how they were arrived at and what they cannot support.
- **Absence of evidence is printed as absence of evidence.** Several topics named in the 2027 syllabus produced no question at all in six papers, and several more are named for the first time in 2027. Each is said so plainly, with no history invented to fill the gap.

Two things the counting found that should change how you study

Two blocks of the paper have barely moved, and they are a third of it. Materials, Manufacturing and Metrology was the largest technical subject in **every one of the six papers**, at 14 to 21 marks, and Engineering Mathematics carried **exactly 13 marks, in nine questions, in all six**. Between them they account for 181 of the 510 technical marks set across the record. Nothing else in the paper is that steady, and a preparation that is thin in either is thin where the paper has been most predictable.

One paper moved three subjects by six marks or more at once. The 2023 paper gave Mechanics of Materials **15** marks, against 6 in both 2022 sets and 6 again in 2024, and gave Theory of Machines and Vibrations and Industrial Engineering **2 marks each**, against 8 in both 2022 sets and 8 again in 2024. Nothing in the other five papers resembles it. A candidate who plans on any one year's shape is planning on one draw from a range that six papers have only begun to show. This book prints the range beside every mean, and the range is the number to plan on.

The two findings point the same way: secure what has held in every paper, and prepare the rest to the width of its range rather than to its latest value. Six papers in five years show patterns — what has been asked, how often, in what form. They do not show trends, and this book does not claim any.

What is in it

The official 2027 syllabus with the 2026-to-2027 comparison generated from the source documents; a chapter on the organising institute and why none of the six counted papers carries its name; the counted weightage of every subject and, inside each subject, of every topic, over all six papers; eleven subject chapters — one for each of the ten technical subjects and one for General Aptitude — with a question bank in every one, each question with a worked solution and four confidence figures; a chapter explaining exactly what those figures mean and what they do not; a hundred-question priority list; a chapter on what not to skip; ten full-length mock tests in the 2027 pattern with keys and worked solutions; a staged revision plan; a last-minute sheet; and a statement of sources and their limits.

Every question is new. None is copied from a past paper, and the mock tests were written under a rule that forbids reusing any question from the subject chapters. What is borrowed from the past is the pattern, and the pattern is what repeats.

Section 1 — Executive Summary

If you read nothing else in this book, read this. Everything here is drawn from the six official GATE Mechanical Engineering papers of the last five years — both sets of 2022 and the single paper of each year from 2023 to 2026 — counted question by question against their official answer keys, and from the official syllabus documents of 2024, 2025, 2026 and 2027. Every figure can be checked; every claim names what it rests on.

Statements are labelled. **FACT** is something counted or quoted. **ANALYSIS** is an inference from it. **PREDICTION** is a statement about a paper that has not been set. They are never blended.

A note on the averages. 2022 was set as two papers and every later year as one, so a plain average over six papers would count 2022 twice. Every **mean** in this book is therefore a mean a year: the two 2022 sets are averaged first, then the five years. Totals ("103 of the 510 technical marks") are plain sums over all six papers.

The eight things that matter most

1. The 2027 syllabus changed, and most of the change is in manufacturing and industrial engineering

FACT. The 2026 and 2027 ME syllabus documents were compared phrase by phrase and word by word by a program. Result: **30 phrases added, 5 removed, 19 reworded**, with 2 more moved to another paragraph. The document grew from 214 phrases to 249. Of the four syllabus sections, Section 4 — Materials, Manufacturing and Industrial Engineering — changed most: 38 of its 2027 phrases are not in 2026, and 18 of its 2026 phrases are not in 2027. Two paragraph labels are new, **Additive Manufacturing** and **Computer Aided Manufacturing and Automation**; **two are gone**, **Computer Integrated Manufacturing** and **Inventory Control**, whose contents moved into their neighbours.

FACT. Only two phrases left the syllabus with words the 2027 document uses nowhere: **Lagrange's equation (Engineering Mechanics)** and **Heisler's charts** (Heat Transfer). Among the additions are these, each carrying at least one word the 2026 document never printed: the **transmissibility ratio** and a **PID controller**, in a new control-systems line under Vibrations; **boiling and condensation**; **basics of combustion**, with air-fuel and equivalence ratios; **pumps and pump characteristics**; **belt drives**; casting defects; pneumatic, electro-pneumatic and hydraulic actuators **and** programmable logic controllers; **and** work study, productivity, six sigma, and quality and reliability concepts.

FACT. The 2024, 2025 and 2026 documents are identical in wording. The only differences among them are three paragraph labels whose capitalisation or hyphen changed between 2024 and 2025.

ANALYSIS. The papers of 2024, 2025 and 2026 were set on a syllabus that did not name the additions above, so no counted question in those three papers can have been set on them under those names. That is the whole meaning of a zero beside a newly named topic: it is not evidence that the topic is rare, only that it is new. The 2022 and 2023 syllabus documents were not obtained for this edition, and this book claims nothing about them.

What to do. Read Section 2 before you drop anything from an older reading list, and give the newly named topics a short, honest reading: past papers cannot prepare you for them.

2. Materials, Manufacturing and Metrology is the largest technical subject in every paper

FACT. Materials, Manufacturing and Metrology carried **15, 17, 21, 16, 20 and 14 marks** in 2022 S1, 2022 S2, 2023, 2024, 2025 and 2026: a mean of **17.4 a year**, and **103 of the 510** technical marks set across the six papers, 20.2 per cent. It was the largest technical subject in all six papers.

ANALYSIS. The one ranking in the record that never changed. It is also the subject whose syllabus changed most in 2027, so its largest block is the one where a candidate's older notes are most likely to be out of date — Section 2.2 lists every phrase.

3. Engineering Mathematics is 13 marks in every paper, and General Aptitude is 15

FACT. Engineering Mathematics carried ****exactly 13 marks**, in exactly nine questions — five at one mark and four at two — in all six papers. **** General Aptitude** carried ten questions, five at one mark and five at two, totalling exactly 15 marks, in all six; all sixty General Aptitude questions set were multiple-choice questions.

ANALYSIS. Together they are 28 of every paper's 100 marks, and neither has moved by a single mark in five years. They are the most predictable part of your score, and General Aptitude alone is worth more than every technical subject except Materials, Manufacturing and Metrology.

What to do. Do not leave either to the last month.

4. IIT Madras sets 2027, and it set none of the six counted papers

FACT. GATE 2027 is organised by IIT Madras. The six counted papers were organised by ****IIT Kharagpur** (2022, both sets), IIT Kanpur (2023), IISc Bengaluru (2024), IIT Roorkee (2025) and IIT Guwahati (2026). ******

IIT Madras last organised GATE in 2019, three years before the earliest paper counted here.

ANALYSIS. There is no institute-matched evidence in this book's record at all. Any claim that "IIT Madras prefers" some kind of ME question is not an inference from anything in it. The one public decision IIT Madras has made about the ME paper is the 2027 syllabus — and Section 2 prints it in full.

5. Six papers in five years show patterns, not trends

FACT. Summing the subject-by-subject differences in technical marks, the two 2022 sets — one institute, one year — differ from each other by **4 marks** out of 85. Papers from different years differ by **4 to 36 marks**, 21.3 on average over the fourteen such pairs. The 2023 paper differs from every other paper by 30 marks or more; without it, papers of different years differ by 4 to 24.

FACT. No technical subject rose in every year or fell in every year. Engineering Mathematics was flat; every other subject moved in both directions at least once.

ANALYSIS. The paper's shape moves between years, sometimes a great deal, and there are only five years.

Nothing in this book is claimed as a trend. Where a subject moved one way over two or three years, it says so and says that five years cannot tell that apart from chance.

What to do. Read the range beside every mean in this book. Your paper gets one value out of that range and you do not choose which.

6. Where the marks are

FACT. Marks by subject in the six papers, counted against the 2027 syllabus, with the mean a year and the range:

Subject	2022 S1	2022 S2	2023	2024	2025	2026	Mean a year	Range
9. Materials, Manufacturing and Metrology	15	17	21	16	20	14	17.4	14-21
General Aptitude	15	15	15	15	15	15	15.0	15-15
1. Engineering Mathematics	13	13	13	13	13	13	13.0	13-13
8. Thermodynamics and Applications	8	8	7	8	11	10	8.8	7-11
3. Mechanics of Materials	6	6	15	6	7	8	8.4	6-15
6. Fluid Mechanics	8	8	10	7	8	6	7.8	6-10
7. Heat Transfer	8	8	7	9	5	8	7.4	5-9
10. Industrial Engineering	8	8	2	8	6	10	6.8	2-10
4. Theory of Machines and Vibrations	8	8	2	8	8	7	6.6	2-8
2. Engineering Mechanics	8	6	6	6	2	6	5.4	2-8
5. Machine Design	3	3	2	4	5	3	3.4	2-5

ANALYSIS. The top three technical subjects — Materials, Manufacturing and Metrology, Engineering Mathematics, and Thermodynamics and Applications — carry 233 of the 510 technical marks, 45.7 per cent.

The widest swings in the record come from a single paper. Mechanics of Materials ran 6 to 8 marks in five papers and 15 in 2023; Theory of Machines and Vibrations ran 7 to 8 in the other five and Industrial Engineering 6 to 10, and both drew 2 in 2023.

7. Over a third of the topics have been examined in five or six papers, and fourteen in none

FACT. Grouping each subject's syllabus into the topic groups its chapter uses gives 86 technical groups.

14 appeared in all six papers, carrying **153 of the 510** technical marks; 17 more appeared in five of the six (174 marks); and **14 appeared in none**. Some of the fourteen are long-standing syllabus lines no paper has set — virtual work, cams, critical speeds of shafts, strain gauges and rosettes, fluid properties, modes of heat transfer. Others are among the 2027 additions — control systems, boiling and condensation, combustion, work study and quality.

ANALYSIS. The 31 groups examined in five or six papers carry 327 of the 510 technical marks, 64.1 per cent. That is the core to secure first. The never-examined groups are of two kinds and should be read differently: a long-named topic no paper has set is a genuine zero over six papers, and it still says nothing about 2027; a newly named topic has a zero because it is new.

What to do. Study the five-and-six-paper topics to depth, and give every never-examined topic a short, honest reading: they are named, most are short, and no past paper can prepare you for them.

8. The multiple-select question has almost gone; the numerical question has not

FACT. MCQ, MSQ and NAT counts over the full 65 questions: **35, 6, 24** in 2022 S1; **35, 6, 24** in 2022 S2; **32, 8, 25** in 2023; **39, 2, 24** in 2024; **38, 1, 26** in 2025; **45, 0, 20** in 2026. Every MSQ and every NAT was in a technical subject. Across the six papers, **143 of the 330 technical questions** were NATs and 23 were MSQs.

FACT. MSQ carries no negative marking and no partial credit: you score only by selecting every correct option and no incorrect one. NAT carries no negative marking, and the answer must fall inside the key's range.

ANALYSIS. The fall in MSQs from 8 in 2023 to none in 2026 is the one run in the record that looks like a direction, and it is a choice of format by the setters of four papers at four institutes, not a change of rule. Five years cannot say whether IIT Madras will continue it. Prepare for a handful of MSQs or none, and for 20 to 26 NATs — the format in which a correct method with careless arithmetic scores nothing.

What this book will not do

It will not tell you it knows the paper. Three commitments follow from that, and they are kept everywhere in this book:

- **Facts, analysis and predictions are labelled separately**, as above.
- **Confidence is reported in four dimensions.** Being confident that the Rankine cycle will be examined is not the same as being confident about a particular reheat-cycle question. Every question in the bank carries Topic, Concept, Question-pattern and Exact-question confidence, and the last is capped, because GATE repeats concepts and not wording.
- **Absence of evidence is printed as absence of evidence.** Several topics named in the 2027 syllabus produced no question in six papers, and several are named for the first time. Each is said so plainly, with no history invented to fill the gap, and questions written on syllabus authority alone say that this is what they rest on.

Additive Manufacturing: Principles of different types of additive manufacturing processes for engineering materials; Additively manufactured products – advantages, limitations, and applications.

Metrology and Inspection: Limits, fits and tolerances; Linear and angular measurements; Comparators; Interferometry; Form and finish measurement; Alignment and testing methods; Tolerance analysis in manufacturing and assembly; Principle of coordinate-measuring machine (CMM).

Computer Aided Manufacturing and Automation: Basic concepts of CAD/CAM and their integration tools; NC/CNC machines and CNC programming; Pneumatic, electro-pneumatic, and hydraulic actuators for automation; Programmable logic controllers.

Production Planning and Control: Work study, Productivity, Forecasting models, Aggregate production planning, Scheduling, Materials requirement planning, Six sigma, Lean manufacturing, Inventory Control – deterministic models, safety stock, inventory control systems, Quality and reliability concepts for product lifecycle analysis.

Operations Research: Linear programming, Simplex method, Transportation, Assignment, Network flow models, Simple queuing models, PERT and CPM.

***Note this, because it decides more marks than any typographical point above.** The word "bending" appears in the 2027 syllabus only in Mechanics of Materials — "bending moment diagrams", "bending and shear stresses" — that is, beam theory. The 2026 document also named **bending** as a sheet metal-forming process; 2027 no longer lists it by name, and says only "sheet forming processes". Likewise "**Plastic deformation and yield criteria**" sits under Casting, Forming and Joining, while "**failure theories**" sits under Machine Design, and a question can belong to either: 2025 (Q22), the von Mises yield stress in pure shear, was counted under the first with the second recorded as its alternative. Every question in this book says which it means.*

2.2 What changed between 2026 and 2027

FACT. The comparison below was made by a program that reads the two official PDFs and compares them phrase by phrase and word by word. No coaching site's "GATE 2027 syllabus changes" article was opened, quoted or checked against: a change list copied from another change list is worth nothing, and a wrong one is worse than none.

The method. Each PDF's text is read and its running page header dropped. Each section is split at its paragraph labels, and each paragraph into phrases on its own commas, semicolons, colons and full stops.

The 2026 and 2027 phrase lists are aligned after folding only case and spacing, so that a change of a single word or letter appears as a reworded phrase. Labels are matched across the two years by name; a phrase that reappears word for word in another paragraph is reported as moved rather than as removed and added. Each paragraph is then compared token by token — every word and every punctuation mark — to catch changes of capitalisation, spelling or punctuation that the phrase comparison does not show. Finally the whole of each document is checked for any word that appears in one and nowhere in the other.

The result, in one line. 30 phrases added, 5 removed, 19 reworded; 2 phrases moved word for word to another paragraph and 2 to another place in the same paragraph; 24 further differences of capitalisation, spelling or punctuation.

By syllabus section. Phrases compared as a whole section, so that a phrase which only moved between two paragraphs of the same section is not counted:

Section	2026 phrases	2027 phrases	Only in 2026	Only in 2027
1. Engineering Mathematics	46	46	2	2
2. Applied Mechanics and Design	53	59	11	17
3. Fluid Mechanics and Thermal Sciences	63	72	6	15

Section	2026 phrases	2027 phrases	Only in 2026	Only in 2027
4. Materials, Manufacturing and Industrial Engineering	52	72	18	38

By paragraph, with this book's subject for each:

Subject	Paragraph (2027)	2026 phrases	2027 phrases	Added	Removed	Reworded
1	Linear Algebra	3	3	0	0	0
1	Calculus	20	20	0	0	0
1	Differential Equations	7	7	0	0	0
1	Complex Variables	4	4	0	0	1
1	Probability and Statistics	9	9	0	0	0
1	Numerical Methods	3	3	0	0	1
2	Engineering Mechanics	14	13	0	1	2
3	Mechanics of Materials	16	18	0	0	3
4	Theory of Machines	8	8	0	0	0
4	Vibrations	5	9	4	0	1
5	Machine Design	10	11	1	0	1
6	Fluid Mechanics	17	19	1	0	1
7	Heat Transfer	19	19	1	1	0
8	Thermodynamics	9	12	0	0	1
8	Applications	18	22	4	0	0
9	Engineering Materials	4	7	0	0	2
9	Casting, Forming and Joining Processes	12	16	1	0	3
9	Machining and Machine Tool Operations	11	11	3	0	2
9	Additive Manufacturing (new label)	0	4	4	0	0
9	Metrology and Inspection	9	9	0	0	1
9	Computer Aided Manufacturing and Automation (new label)	0	6	4	0	0
10	Production Planning and Control	5	12	7	0	0
10	Operations Research	7	7	0	0	0
—	Computer Integrated Manufacturing (2026 only)	2	0	0	1	0
—	Inventory Control (2026 only)	2	0	0	2	0
	Total	214	249	30	5	19

Five paragraphs are unchanged word for word and mark for mark: Linear Algebra, Calculus, Differential Equations, Probability and Statistics, and Theory of Machines. Every one of the ten subjects has at least one paragraph that changed.

The removals

Five phrases printed in 2026 are not printed in 2027. Only two of them have gone from the syllabus:

Section 3 — The Organising Institute, and the Six Papers It Did Not Set

GATE 2027 is organised by IIT Madras.

Each year one institute takes responsibility for the examination, and the paper is set under its supervision. The six Mechanical Engineering papers counted for this book were set under five different institutes, and **none of them was IIT Madras**. There is no paper in this book's record whose setter is the setter of the paper you are preparing for.

This chapter says so, says what that rules out, and then says what the six papers set under other institutes do show — the features of the paper that survived four changes of organiser. It is a short chapter because the honest version of it is short. A chapter that found an "IIT Madras style" in papers IIT

Madras did not set would be inventing it.

3.1 What is fact here, and what is not

FACT. GATE 2027 is organised by IIT Madras. The official 2027 ME syllabus document carries the header "IIT Madras | Organizing Institute", and the official portal is gate2027.iitm.ac.in.

FACT. The organising institutes of the six counted ME papers are: **IIT Kharagpur (2022, both sets), IIT Kanpur (2023), IISc Bengaluru (2024), IIT Roorkee (2025) and IIT Guwahati (2026).**

FACT. IIT Madras last organised GATE in 2019, three years before the earliest paper counted here. The 2019 ME papers are not part of this book's record.

NOT A FACT, and this book will not pretend otherwise. That an institute "has a style" which carries over into a paper it did not set, or across a gap of years and a change of everyone involved. Papers are set by subject panels, not by buildings. Whatever IIT Madras's panels did in 2019, they did it on a syllabus older than any this book holds, for papers this book did not count.

3.2 Six papers, five institutes

FACT. Technical marks by subject, each paper counted question by question against its official key:

Subject	2022 S1, IIT Kharagpur	2022 S2, IIT Kharagpur	2023, IIT Kanpur	2024, IISc Bengaluru	2025, IIT Roorkee	2026, IIT Guwahati	Mean a year
1. Engineering Mathematics	13	13	13	13	13	13	13.0
2. Engineering Mechanics	8	6	6	6	2	6	5.4
3. Mechanics of Materials	6	6	15	6	7	8	8.4
4. Theory of Machines and Vibrations	8	8	2	8	8	7	6.6
5. Machine Design	3	3	2	4	5	3	3.4
6. Fluid Mechanics	8	8	10	7	8	6	7.8
7. Heat Transfer	8	8	7	9	5	8	7.4
8. Thermodynamics and Applications	8	8	7	8	11	10	8.8
9. Materials, Manufacturing and Metrology	15	17	21	16	20	14	17.4
10. Industrial Engineering	8	8	2	8	6	10	6.8
General Aptitude	15	15	15	15	15	15	15.0

ANALYSIS. For every year but one, the organiser and the year are the same variable. IIT Kanpur set exactly one ME paper in this record, in exactly one year, so there is no way to tell whether 2023's 15 marks of Mechanics of Materials and 2 each of Theory of Machines and Vibrations and Industrial Engineering reflect IIT Kanpur's panel, that year's

plan, or neither. The same is true of IIT Roorkee's 2 marks of

Engineering Mechanics in 2025 and IIT Guwahati's 10 of Industrial Engineering in 2026. **Any claim about what a particular institute "does" in ME rests on one year and cannot be separated from chance.** This book makes none.

FACT. The one exception is 2022, when IIT Kharagpur set two papers. Summing the subject-by-subject differences, its two sets differ from each other by **4 marks** out of 85 — Engineering Mechanics 8 against

6, Materials, Manufacturing and Metrology 15 against 17, and every other subject identical. Papers from different years differ by 4 to 36 marks (Section 4.4).

ANALYSIS. One pair of sets is one observation. It is consistent with the two sets of a year being drawn to a single plan, and it cannot show whether that plan is the institute's or the year's.

3.3 What IIT Madras has already decided about ME

FACT. The one ME document IIT Madras has published so far is the 2027 syllabus. Its PDF metadata gives a creation date of 6 July 2026. Compared phrase by phrase with the 2026 document published by IIT

Guwahati, it adds 30 phrases, removes 5 and rewords 19, creates the paragraphs **Additive Manufacturing** and

Computer Aided Manufacturing and Automation, and folds **Computer Integrated Manufacturing** and

Inventory Control into their neighbours (Section 2.2). The three documents before it — 2024, 2025 and

2026, from three different institutes — had changed nothing between them but the capitalisation or hyphenation of three paragraph labels.

ANALYSIS. That is the only public decision about the ME paper that can be attributed to the 2027 organiser, and it is a real one: the first change to the wording of the ME syllabus in the four documents this book holds. What it adds is known exactly — control systems under Vibrations, boiling and condensation, combustion, pumps, actuators and programmable logic controllers, work study, productivity, six sigma, and quality and reliability among it. What it does not say is how many marks the additions will carry, and nothing in the record can say that either, because no counted paper was set on them. It supports reading every added line. It supports no forecast of their weight.

3.4 What survived four changes of organiser

If one institute's paper cannot tell you about IIT Madras, five institutes' papers can at least tell you what did not change when the institute did.

FACT. In all six papers, set under five different institutes:

Feature	2022 S1	2022 S2	2023	2024	2025	2026
Questions / marks	65 / 100	65 / 100	65 / 100	65 / 100	65 / 100	65 / 100
One-mark / two-mark questions	30 / 35	30 / 35	30 / 35	30 / 35	30 / 35	30 / 35
General Aptitude	10 questions, 15 marks, all MCQ	the same	the same	the same	the same	the same
Block layout	Q1–5 and Q11–35 one mark; Q6–10 and Q36–65 two marks	the same	the same	the same	the same	the same
Engineering Mathematics	9 questions, 13 marks	the same	the same	the same	the same	the same
MCQ / MSQ / NAT	35 / 6 / 24	35 / 6 / 24	32 / 8 / 25	39 / 2 / 24	38 / 1 / 26	45 / 0 / 20
Largest technical subject	Materials, Manufacturing and Metrology (15)	the same (17)	the same (21)	the same (16)	the same (20)	the same (14)

Section 4 — What Six Papers Actually Contain

This is the chapter the rest of the book rests on. Everything after it — the subject chapters, the question banks, the mock tests — is built on the counts below, so this chapter shows you the counts, the method that produced them, and the places where the method needed a judgement.

It is also the chapter that has to say most plainly how much six papers can prove. **The record is the five most recent years of GATE Mechanical Engineering: two papers in 2022, one in each year from 2023 to 2026.** Six papers show patterns — what has been asked, how often, in what form. They do not show trends, and this chapter does not claim any.

4.1 How the counting was done

FACT: All six official GATE Mechanical Engineering papers of 2022 to 2026 were read one question at a time: 2022 Set 1, 2022 Set 2, 2023, 2024, 2025 and 2026, **390 questions and 600 marks.**

Each question was read from the organising institute's own question paper — not from a key, not from a summary, not from a solved-papers book. Its marks, its question type and its answer were then taken from **the official answer key for that paper** and from nothing else. So when this book says a question carried two marks and was a numerical-answer question, that is the organisers' record of it and not a reading of the paper's layout.

Each question was then assigned to one of this book's ten technical subjects, and to the paragraph of the **2027** syllabus it falls under, and to a topic inside it, in the syllabus's own words wherever the syllabus has words for it. That is the point of the exercise: a 2027 candidate needs to know where the marks would land under the syllabus they will actually sit, and Section 2 shows that it is not the syllabus the papers were set on.

Every count had to close, or it was not a count. Each of the six closes on:

Check	Required	Result
Questions	65	65 in all six
Total marks	100	100 in all six
One-mark questions	30	30 in all six
Two-mark questions	35	35 in all six
General Aptitude	exactly 15	15 in all six (10 questions)
Technical	exactly 85	85 in all six
Marks and type of every row	as in the official key	match in all six

A count that did not close would have been re-done rather than adjusted. Nothing was moved, merged or dropped to make a total come out right.

All six papers use the same layout, and the count confirms it question by question: Q1 to Q10 are General Aptitude, Q1 to Q5 at one mark and Q6 to Q10 at two; the technical questions run from Q11 to Q65, Q11 to Q35 at one mark and Q36 to Q65 at two. A citation such as "2025 (Q35)" in this book is the paper's own printed number. For 2022 the set is always named — "2022 S1 (Q35)", "2022 S2 (Q35)" — because the number alone would fit two questions.

Where a question could honestly sit in two subjects, both are recorded. The question is placed by the boundary rules stated in Section 2.4, the other candidate subject is recorded beside it, and where no rule settles the matter the question is marked as counted at its lean, with the reason written out. **52 of the

330 technical questions** carry an alternative in another subject, and thirteen questions are counted at a lean. Section 4.7 prints the thirteen and shows what moving them would do.

How the averages are taken. 2022 had two papers and every other year one. A plain mean over six papers would count 2022 twice, so every **mean** in this book is a mean a year: the two 2022 sets are averaged first, then the five years. Totals and shares are plain sums over all six papers.

Why not use a published weightage table. Because a table that does not print its boundary decisions cannot be checked, and in ME the boundaries decide real marks: 78 of the 510 technical marks set in these six papers sit on questions that could have been counted in a second subject. A count that names every question it placed and where else it could have gone is a different kind of object from a table of totals.

4.2 The papers

FACT.

Paper	Organising institute	Session recorded in the key	Question paper	Answer key
2026	IIT Guwahati	Session 6	counted	official key
2025	IIT Roorkee	Session 3	counted	official key
2024	IISc Bengaluru	Session 2	counted	official final key
2023	IIT Kanpur	Session 2	counted	official key
2022 S2	IIT Kharagpur	Session 8	counted	official key
2022 S1	IIT Kharagpur	Session 7	counted	official key

Three facts about this base that you should have rather than discover:

- **It stops at 2022.** Earlier ME papers exist and are not counted here, and this book cites none of them. Five years is the whole of the record, and every conclusion below is sized to it.
- **2022 is two papers; every later year is one.** The 2022 file holds Set 1 and Set 2, each with its own key; from 2023 the organising sites list a single ME paper a year, so a year and a question number identify a question completely.
- **Seven questions were awarded to everyone.** The official keys mark 2022 S1 (Q10), 2023 (Q55) and 2024 (Q7, Q10, Q19, Q34, Q63) as MTA, marks to all. Each is counted, with the marks it carried and in the subject it was set as, because the paper awarded them. Section 3.5 lists them with the keys that accept more than one answer.

4.3 Marks by subject, every paper

FACT. Technical marks by subject. General Aptitude is 15 in every paper and is listed last.

Subject	2022 S1	2022 S2	2023	2024	2025	2026	Mean a year	Range	Total	Share of 510 technical marks
9. Materials, Manufacturing and Metrology	15	17	21	16	20	14	17.4	14-21	103	20.2%
1. Engineering Mathematics	13	13	13	13	13	13	13.0	13-13	78	15.3%
8. Thermodynamics and Applications	8	8	7	8	11	10	8.8	7-11	52	10.2%
3. Mechanics of Materials	6	6	15	6	7	8	8.4	6-15	48	9.4%
6. Fluid Mechanics	8	8	10	7	8	6	7.8	6-10	47	9.2%
7. Heat Transfer	8	8	7	9	5	8	7.4	5-9	45	8.8%
10. Industrial Engineering	8	8	2	8	6	10	6.8	2-10	42	8.2%
4. Theory of Machines and Vibrations	8	8	2	8	8	7	6.6	2-8	41	8.0%
2. Engineering Mechanics	8	6	6	6	2	6	5.4	2-8	34	6.7%
5. Machine Design	3	3	2	4	5	3	3.4	2-5	20	3.9%
Outside the 2027 syllabus	0	0	0	0	0	0	<i>0.0</i>	0-0	0	0%
General Aptitude	15	15	15	15	15	15	15.0	15-15	90	—

Concept Tested: The affinity laws for one pump at two speeds: $Q \propto N$, $H \propto N^2$, $P \propto N^3$.

Why This Question Is High Probability, and the honest caveat: "pumps and pump characteristics" is ****new** in the 2027 syllabus^{**}; no counted paper could have asked it, so the confidence figures rest on the syllabus alone. The affinity laws are the simplest quantitative content of the line.

SOLUTION

For the same pump (same diameter) at homologous points, the flow coefficient $Q/(ND^3)$ and the head coefficient $gH/(N^2D^2)$ are unchanged, so with D fixed,

$$H_2 = H_1 (N_2/N_1)^2 = 30 \times (1740/1450)^2 = 30 \times 1.2^2 = \mathbf{43.2 \text{ m}}. \text{ Answer (B).}$$

The other two laws give $Q_2 = 0.05 \times 1.2 = 0.06 \text{ m}^3/\text{s}$ and $P_2 = 18 \times 1.2^3 = 31.1 \text{ kW}$.

Why the squares and cubes. The head comes from the velocity the impeller gives the water; and a velocity head goes as the square of the tip speed; the power is flow $\times$ head, so it goes as $N \times N^2 = N^3$.

The traps. (A) 36.0 m scales the head linearly, the law for flow. (C) 51.8 m scales it with the cube, the law for power. (D) 30.0 m assumes the head is set by the impeller diameter alone and does not change with speed.

5.9 Materials, Manufacturing and Metrology

Materials, Manufacturing and Metrology carries **103 marks across the six counted papers**: 68 questions, a mean of 17.4 marks a year, and never fewer than 14 in any paper. That is ****20.2 per cent of the 510 technical marks**** the six papers set, and it makes this the largest single block in the ME paper, ahead of General Aptitude (15 a paper) and Engineering Mathematics (13). The lowest paper, 2026, gave it 14 marks; the highest, 2023, gave it 21.

It is also the widest section in the syllabus: six paragraphs, from crystal structures to programmable logic controllers, and the 2027 syllabus rewrote more of it than of any other part of the paper. Two paragraph labels are new (Additive Manufacturing, and Computer Aided Manufacturing and Automation), and several topics are named for the first time: casting defects, solid-state welding, non-destructive testing, ASA and ORS tool geometry, actuators and PLCs. Every citation below names the paper and the question number, and every figure-based question is restated in words, because this book has no drawings.

The official 2027 syllabus for this section

****FACT — GATE 2027 ME, Section 4: Materials, Manufacturing and Industrial Engineering — Materials, Manufacturing and Metrology**** (official syllabus PDF, IIT Madras)

Engineering Materials: Crystal structure of common metals, phase diagrams, heat treatment; Properties and applications of engineering materials - metals and alloys, polymers, composites and ceramics; Engineering and true stress-strain diagrams.

Casting, Forming and Joining Processes: Different types of casting processes; Patterns, moulds and core; Solidification and cooling, riser and gating design, casting defects; Plastic deformation and yield criteria; Hot working and cold working; Principles of bulk forming and sheet forming processes and estimation of load in these processes; Fundamentals of powder metallurgy and its applications; Principles of welding, brazing, soldering; Solid-state welding processes, adhesive bonding; Non-destructive testing methods.

Machining and Machine Tool Operations: Basic machining operations and machine tools; Single-point and multipoint cutting tools; Geometry of single-point turning tools - ASA and ORS systems; Mechanics of machining; Cutting tool materials, tool life and tool wear; Economics of machining; Abrasive machining processes; Principles and applications of non-traditional machining processes; Principles of work holding, jigs and fixtures.

Additive Manufacturing: Principles of different types of additive manufacturing processes for engineering materials; Additively manufactured products – advantages, limitations, and applications.

Metrology and Inspection: Limits, fits and tolerances; Linear and angular measurements; Comparators; Interferometry; Form and finish measurement; Alignment and testing methods; Tolerance analysis in manufacturing and assembly; Principle of coordinate-measuring machine (CMM).

Computer Aided Manufacturing and Automation: Basic concepts of CAD/CAM and their integration tools; NC/CNC machines and CNC programming; Pneumatic, electro-pneumatic, and hydraulic actuators for automation; Programmable logic controllers.

Two notes on that text. "coordinate-measuring" carries a space after the hyphen, and "Patterns, moulds and core" has a singular "core" after two plurals; both are the PDF's own and are reproduced, not tidied. The other two paragraphs of Section 4, Production Planning and Control and Operations Research, belong to Section 5.10.

5.9.1 What the 2027 syllabus changed here

FACT — Engineering Materials: two phrases reworded. 2026 read "Structure and properties of engineering materials, phase diagrams, heat treatment, stress-strain diagrams for engineering materials." 2027 reads "Crystal structure of common metals, phase diagrams, heat treatment; Properties and applications of engineering materials - metals and alloys, polymers, composites and ceramics; Engineering and true stress-strain diagrams."

FACT — Casting, Forming and Joining Processes: one phrase added, three reworded. 2026 read "Different types of castings, design of patterns, moulds and cores; solidification and cooling; riser and gating design. Plastic deformation and yield criteria; fundamentals of hot and cold working processes; load estimation for bulk (forging, rolling, extrusion, drawing) and sheet (shearing, deep drawing, bending) metal forming processes; principles of powder metallurgy. Principles of welding, brazing, soldering and adhesive bonding." 2027 adds "0"; turns "Different types of castings, design of patterns, moulds and cores" into "Different types of casting processes; Patterns, moulds and core"; replaces the bracketed process lists with "Principles of bulk forming and sheet forming processes and estimation of load in these processes" and adds "and its 0" to powder metallurgy; and turns "soldering and adhesive bonding" into "soldering; Solid-state welding processes, adhesive bonding; Non-destructive testing methods."

FACT — Machining and Machine Tool Operations: three phrases added, two reworded, one moved out. 2026 read "Mechanics of machining; basic machine tools; single and multi-point cutting tools, tool geometry and materials, tool life and wear; economics of machining; principles of non-traditional machining processes; principles of work holding, jigs and fixtures; abrasive machining processes; NC/CNC machines and CNC programming." 2027 adds "0 and machine tools", "Single-point and multipoint cutting tools" and "0"; rewords the tool line to "Cutting tool materials, tool life and tool wear" and the non-traditional line to "Principles 0 of non-traditional machining processes"; and moves "NC/CNC machines and CNC programming" word for word to the new automation paragraph.

FACT — two new paragraph labels. 2026 had a paragraph "Computer Integrated Manufacturing: Basic concepts of CAD/CAM and their integration tools; additive manufacturing." 2027 has no such label. Its CAD/CAM phrase moves word for word to the new **Computer Aided Manufacturing and Automation** paragraph, joined there by the NC/CNC phrase and by two added lines, "Pneumatic, electro-pneumatic, and hydraulic actuators for automation" and "0". Its bare phrase "additive manufacturing" becomes a paragraph of its own, **Additive Manufacturing**, which adds "Principles of different types of additive manufacturing processes for engineering materials" and "Additively manufactured products – 0".

FACT — Metrology and Inspection: one phrase reworded. "concepts of coordinate-measuring machine (CMM)" became "Principle of coordinate-measuring machine (CMM)"; every other change in the paragraph is a capital letter.

Metrology and Inspection — 12 marks, 12 per cent

FACT. Eight questions: 2022 S1 (Q20) the type of fit from shaft and hole limits; 2022 S2 (Q20) allowance, maximum clearance and maximum material limit; 2022 S2 (Q56) the best-size wire for a metric thread; 2023 (Q53) the six-sigma upper limit of a derived dimension; 2023 (Q54) the distance between two CMM-fitted planes; 2023 (Q56) the height difference of slip gauges from interference fringes; 2025 (Q25) the fit for a given hole and shaft; 2026 (Q33) the ten-point height of a surface trace.

ANALYSIS. Limits and fits appeared three times, always as reading limits; the other five are one each of thread measurement, tolerance stacking, CMM, interferometry and surface finish. Comparators and alignment testing have not been examined.

Computer Aided Manufacturing and Automation — 8 marks, 8 per cent

FACT. Six questions: 2022 S1 (Q19) the stepper-motor pulse rate for a lead-screw table; 2022 S2 (Q22) which drive cannot give linear motion (chain and sprocket); 2023 (Q45) the controllable positions of a 50-step lead-screw axis; 2024 (Q33) the letter for preparatory functions (G); 2025 (Q23) which representation builds solid models (B-rep); 2026 (Q45) which polyhedron violates Euler's law.

ANALYSIS. Something from this paragraph came in every paper. The stepper-motor resolution calculation came twice. Pneumatic, electro-pneumatic and hydraulic actuators and PLCs are new and have no precedent.

5.9.4 How to prepare this section

1. **Mechanics of machining first.** Every paper had a 2-mark question on it. Drill the Merchant circle (shear angle from the chip ratio, F_s , F_c , the friction and normal forces), then Taylor's equation in both forms (two speed-life pairs, and life counted in parts or holes), then the minimum-cost tool life.
2. **Casting, forming and joining as three short formula sets.** Chvorinov's modulus and riser sizing; μ^2R , the limiting drawing strain and blanking clearances; the welding heat balance and the source characteristic. These gave most of the 36 marks.
3. **Materials calculations.** Unit-cell density, the lever rule both ways round, and $\epsilon = n$ at necking. Then read the heat-treatment processes once, with temperatures.
4. **Matching tables for AM, non-traditional machining and casting defects.** Make one table per topic (process, energy, feedstock, application). About two-thirds of the section's 1-mark questions were matching or recognition items.
5. **Metrology and CNC.** Limits and fits until reading them is automatic; the fringe formula; the statistical tolerance stack; stepper resolution; G and M codes.
6. **The new 2027 lines last, but not skipped.** NDT, solid-state welding, ASA/ORS tool signatures, polymers and ceramics, actuators and PLCs are short and descriptive. One sitting covers them.

ANALYSIS — the judgement behind that order. The order follows the counted marks (machining mechanics 18, forming 14, materials 14, casting 12, metrology 12, joining 10) and the fact that the calculations are fewer and more repetitive than the section's breadth suggests: about a dozen formulas carry the numerical marks. The new lines have no evidence either way and are ranked last on that ground alone.

Question Bank — Materials, Manufacturing and Metrology

Forty questions, written for this book, in rough proportion to the counted evidence: five on Engineering Materials; fourteen on Casting, Forming and Joining Processes (four on casting, six on forming, three on joining and one on non-destructive testing, which is new and says so); ten on Machining (six on mechanics, tools, tool life and economics, four on operations, non-traditional machining and work holding); three on

Additive Manufacturing; four on Metrology; and four on CAM and Automation, two of them on the new actuator and PLC lines. Nineteen are MCQ, three MSQ and eighteen NAT; nineteen carry one mark and twenty-one two. Every numerical answer was computed.

QMF.1

NAT · 2 marks

Easy–Moderate

Subject: Materials, Manufacturing and Metrology | **Topic:** Crystal structure of common metals | **Subtopic:** Density of a BCC metal from its atomic radius

Prediction Confidence: Topic 84% | Concept 74% | Question-pattern 64% | Exact-question 16%

Based on PYQs: GATE 2026 (Q62), 2023 (Q51) · last tested 2026 · new question on the same concept, not a reprint

QUESTION

Alpha-iron has a body-centred cubic (BCC) crystal structure. Its atomic radius is 0.124 nm and its atomic weight is 55.85 g/mol. Taking Avogadro's number as 6.022×10^{23} atoms/mol, the theoretical density of alpha-iron is _____ kg/m³ (round off to the nearest integer).

ANSWER 7899

Acceptance band: 7880 to 7915.

Concept Tested: Atoms per BCC cell, the body-diagonal relation between the lattice parameter and the atomic radius, and $\rho = nA/(N_A a^3)$.

Why This Question Is High Probability: "Crystal structure of common metals" is newly worded in 2027, and the unit-cell calculation it names was set as a 2-mark NAT in 2023 (Q51) and again in 2026 (Q62), both for FCC.

SOLUTION

Step 1 — atoms per cell. A BCC cell has one atom at each of eight corners, each shared by eight cells, and one whole atom at the body centre: $n = 8 \times 1/8 + 1 = 2$.

Step 2 — lattice parameter. In BCC the atoms touch along the body diagonal, which spans $r + 2r + r$:

$$\sqrt{3} a = 4r, a = 4 \times 0.124 / \sqrt{3} = \mathbf{0.28637 \text{ nm}} = 2.8637 \times 10^{-10} \text{ m}$$

Step 3 — density. The mass of the cell is nA/N_A and its volume a^3 :

$$\rho = nA/(N_A a^3) = (2 \times 55.85 \times 10^{-3}) / (6.022 \times 10^{23} \times (2.8637 \times 10^{-10})^3) \\ = 0.1117 / (6.022 \times 10^{23} \times 2.3484 \times 10^{-29}) = \mathbf{7899 \text{ kg/m}^3}$$

The two traps. Using the FCC relation $a = 2\sqrt{2} r$ (face diagonal) gives $a = 0.3507 \text{ nm}$ and, with n still 2, a density of 4299 kg/m³, far below any steel. Using $n = 4$ with the BCC lattice parameter doubles the answer.

Keep the pairs together: **SC** $n = 1$, $a = 2r$; **BCC** $n = 2$, $a = 4r/\sqrt{3}$; **FCC** $n = 4$, $a = 2\sqrt{2} r$. The packing factors follow from the same pairs: 0.52, 0.68 and 0.74.

Units. Keep A in kg/mol and a in metres, and the answer is in kg/m³ without further conversion.

QMF.2

MCQ · 2 marks

Moderate

Subject: Materials, Manufacturing and Metrology | **Topic:** phase diagrams | **Subtopic:** Lever rule for pro-eutectoid cementite in a hypereutectoid steel

Prediction Confidence: Topic 84% | Concept 76% | Question-pattern 64% | Exact-question 16%

Based on PYQs: GATE 2024 (Q28), 2023 (Q52), 2022 S1 (Q41) · last tested 2024 · new question on the same concept, not a reprint

QUESTION

A plain carbon steel containing 1.2 wt% carbon is cooled slowly from the austenite region to just below the eutectoid temperature. Take the eutectoid composition as 0.76 wt% C, the carbon content of cementite (Fe₃C) as 6.67 wt% C and the maximum solubility of carbon in ferrite as 0.022 wt% C. The mass fraction of **pro-eutectoid cementite** in the microstructure is closest to

- (A) 0.177
- (B) 0.420
- (C) 0.074
- (D) 0.926

ANSWER (C)

Concept Tested: Which two phase boundaries bound the lever when the pro-eutectoid phase forms, and which arm of the lever gives which phase.

Why This Question Is High Probability: The lever rule on the iron-carbon diagram was set in 2022 S1 (Q41) for pro-eutectoid ferrite and in 2023 (Q52) for pro-eutectoid cementite, both for 2 marks, and 2024 (Q28) asked the phases in pearlite.

SOLUTION

Pro-eutectoid cementite forms from austenite between the A_{cm} line and the eutectoid temperature. Just above the eutectoid, the steel is **pro-eutectoid Fe_3C plus austenite of eutectoid composition**, and all of that austenite becomes pearlite just below it. So the lever runs from 0.76 to 6.67 wt% C, and the fraction of cementite is the arm on the **far** side from cementite:

$$W(\text{pro-eutectoid } Fe_3C) = (1.2 - 0.76)/(6.67 - 0.76) = 0.44/5.91 = \mathbf{0.0745}$$

Answer (C).

Where each wrong option comes from.

- **(A)** is the **total** cementite (pro-eutectoid plus the cementite inside pearlite), from the lever between ferrite and cementite: $(1.2 - 0.022)/(6.67 - 0.022) = 0.177$. The mistake is using the ferrite boundary for a phase that formed above the eutectoid.
- **(B)** is the ratio of the two, $0.0745/0.177 = 0.420$, which is what 2023 (Q52) asked; the mistake is answering a different question from the one set.
- **(D)** is the fraction of **pearlite**, $(6.67 - 1.2)/(6.67 - 0.76) = 0.926$; the mistake is taking the arm on the cementite side, the classic inversion of the lever.

A quick check on any lever answer: a composition close to one end of the lever must give mostly the phase at that end. 1.2 per cent sits near 0.76, so the structure is mostly pearlite and only a little pro-eutectoid cementite, a network at the old austenite grain boundaries.

QMF.3

MSQ · 1 mark

Moderate

Subject: Materials, Manufacturing and Metrology | **Topic:** heat treatment | **Subtopic:** Nitriding, tempering, normalising and the martensite transformation

Prediction Confidence: Topic 84% | Concept 72% | Question-pattern 56% | Exact-question 13%

Based on PYQs: GATE 2025 (Q20), 2022 S1 (Q27) · last tested 2025 · new question on the same concept, not a reprint

QUESTION

Which of the following statement(s) about the heat treatment of steels is/are TRUE?

- (A) Nitriding is carried out below the lower critical (eutectoid) temperature, and the part is not quenched afterwards.
- (B) Tempering a quenched steel raises its hardness above that of the as-quenched martensite.
- (C) Normalising cools the steel in still air, which gives finer pearlite and a higher hardness than full annealing.
- (D) Martensite forms from austenite by the diffusion of carbon atoms out of the lattice during slow cooling.

ANSWER (A), (C)

Concept Tested: What each heat treatment does to the microstructure, and the diffusionless nature of the martensite transformation.

Why This Question Is High Probability: Heat treatment has come as a 1-mark item twice: 2022 S1 (Q27), an MSQ on surface

Section 9 — The Hundred Questions to Work First

If you have time for a hundred questions and not for 292, these are the hundred. If you have time for all 292, work these first anyway, because finishing them leaves you in a defensible position if the time runs out.

9.1 How the hundred were chosen

This list is not hand-picked. One rule was applied to all 292 questions in the eleven subject chapters and the result printed. The rule is stated here so that you can disagree with it precisely.

How many come from each section. Each section's mean marks a year across the five counted years, rounded so that the quotas sum to exactly 100. GATE 2022 was set in two sets, which are averaged first, so 2022 counts once like every other year.

Section	Mean marks a year	Questions in the hundred
Materials, Manufacturing and Metrology	17.4	18
General Aptitude	15.0	15
Engineering Mathematics	13.0	13
Thermodynamics and Applications	8.8	9
Mechanics of Materials	8.4	8
Fluid Mechanics	7.8	8
Heat Transfer	7.4	7
Industrial Engineering	6.8	7
Theory of Machines and Vibrations	6.6	7
Engineering Mechanics	5.4	5
Machine Design	3.4	3

So the hundred mirror the paper. An hour spent on this list is divided the way the marks have been. The means rest on six papers, and a section's marks moved by several between papers, so read each quota as a reasonable share rather than a forecast.

Which questions within a section. Topic confidence highest first — of the four confidence figures this is the one with the most evidence behind it, and Section 7 explains what it is and is not. Ties break on Question-pattern confidence, because a stable question *shape* is a skill that practice transfers to. Any remaining ties break on question number, so the order does not shuffle between printings.

Nothing else enters. Difficulty does not, because an easy question on a near-certain topic is worth more than a hard one on an unlikely topic, and this list is about return on an hour rather than about challenge.

Why the quota is there. Ranked on confidence alone, with no quota, the same rule would have taken 28 from General Aptitude, 19 from Engineering Mathematics, 16 from Heat Transfer, 9 from Materials, Manufacturing and Metrology, 8 from Theory of Machines and Vibrations, 7 from Industrial Engineering, 5 from Mechanics of Materials, 4 from Engineering Mechanics and 4 from Fluid Mechanics. That would have short-changed Fluid Mechanics (4 against a quota of 8), Thermodynamics and Applications (0 against a quota of 9) and Materials, Manufacturing and Metrology (9 against a quota of 18) — not because those sections matter less, but because fewer of their questions carry the very highest Topic figures. A study list that under-weights a section the paper reliably examines is wrong however cleanly it was derived, so the quota decides how many and the confidence figures decide which.

9.2 The blind spot in the rule, stated before the table

The rule ranks on Topic confidence, and Topic confidence rests mostly on the counted record. A topic that none of the six papers has set has **no counted history at all, and so some of the lowest confidence figures in the book**, and its questions drop to the bottom of this list or off it. There are two kinds.

Some the syllabus has long named and no paper has yet set — virtual work, cams and critical speeds of shafts. Others the 2027 syllabus names for the first time, so no counted paper could have been set on them in those words — boiling and condensation, PID control and programmable logic controllers among them; Section 10.2 lists them all.

That is backwards as a study instruction. A never-examined topic is exactly the material no amount of past-paper practice will make you meet, so it is the material most likely to be missing from your preparation on the day. Section 9.4 lists those questions separately, together with every other question whose concept the counted record has never examined. **Work the hundred, then work Section 9.4.** Do not let the ranking decide that a named topic is unimportant; it decided that only because six papers have not yet set it.

9.3 The hundred

Mk is the question's marks. **T** and **P** are its Topic and Question-pattern confidence. Work the question from the subject chapter, not from this table — the table is an index, and the chapters hold the solutions, the acceptance bands and the reasoning.

#	Question	Section	Topic — subtopic	Type	Mk	T	P
1	QMF.20	Materials, Manufacturing and Metrology	Mechanics of machining — Cutting force by the Merchant circle with Merchant's shear angle	NAT	2	88%	68%
2	QMF.23	Materials, Manufacturing and Metrology	Cutting tool materials, tool life and tool wear — Taylor exponent from the number of holes drilled at two speeds	NAT	2	88%	66%
3	QMF.21	Materials, Manufacturing and Metrology	Mechanics of machining — Coefficient of friction on the rake face from measured forces	MCQ	2	88%	64%
4	QMF.22	Materials, Manufacturing and Metrology	Mechanics of machining — Cutting power from specific cutting energy in turning	NAT	1	88%	64%
5	QMF.24	Materials, Manufacturing and Metrology	Economics of machining — Cutting speed for minimum cost per piece	NAT	2	86%	58%
6	QMF.25	Materials, Manufacturing and Metrology	Geometry of single-point turning tools - ASA and ORS systems — Converting an ASA side cutting edge angle to the ORS principal cutting edge angle	MCQ	1	86%	46%
7	QMF.1	Materials, Manufacturing and Metrology	Crystal structure of common metals — Density of a BCC metal from its atomic radius	NAT	2	84%	64%
8	QMF.2	Materials, Manufacturing and Metrology	phase diagrams — Lever rule for pro-eutectoid cementite in a hypereutectoid steel	MCQ	2	84%	64%
9	QMF.7	Materials, Manufacturing and Metrology	riser and gating design — Diameter of a cylindrical riser by Chvorinov's rule	NAT	2	84%	64%
10	QMF.4	Materials, Manufacturing and Metrology	Engineering and true stress-strain diagrams — Engineering UTS from a Hollomon flow curve	NAT	2	84%	58%
11	QMF.9	Materials, Manufacturing and Metrology	casting defects — Cold shut against misrun, hot tear and blowhole	MCQ	1	84%	58%
12	QMF.3	Materials, Manufacturing and Metrology	heat treatment — Nitriding, tempering, normalising and the martensite transformation	MSQ	1	84%	56%
13	QMF.6	Materials, Manufacturing and Metrology	Solidification and cooling — Pattern allowance for solid contraction when a riser feeds solidification shrinkage	NAT	1	84%	56%
14	QMF.8	Materials, Manufacturing and Metrology	riser and gating design — Filling time for top and bottom gating	MCQ	2	84%	48%
15	QMF.11	Materials, Manufacturing and Metrology	Principles of bulk forming and sheet forming processes and estimation of load in these processes — Maximum draft and roll-separating force in flat rolling	NAT	2	82%	62%
16	QMF.13	Materials, Manufacturing and Metrology	Principles of bulk forming and sheet forming processes and estimation of load in these processes — Punch and die sizes for blanking and piercing	MCQ	1	82%	62%
17	QMF.10	Materials, Manufacturing and Metrology	Plastic deformation and yield criteria — Plane-strain flow stress by von Mises	MCQ	1	82%	58%
18	QMF.12	Materials, Manufacturing and Metrology	Principles of bulk forming and sheet forming processes and estimation of load in these processes — Limiting reduction per pass in wire drawing of a strain-hardening metal	MCQ	2	82%	58%
19	QGA.10	General Aptitude	Quantitative aptitude — Probability of exactly one success in three trials	MCQ	1	90%	58%
20	QGA.7	General Aptitude	Quantitative aptitude — Averages worked through totals	MCQ	1	90%	54%
21	QGA.8	General Aptitude	Quantitative aptitude — Average speed over unequal distances	MCQ	1	90%	54%
22	QGA.14	General Aptitude	Quantitative aptitude — Bounds on the probability of two events together	MCQ	2	90%	54%
23	QGA.12	General Aptitude	Quantitative aptitude — Mensuration — circle and square of equal perimeter	MCQ	2	90%	52%
24	QGA.16	General Aptitude	Quantitative aptitude — Geometry — a square inside a circle, and a circle inside that square	MCQ	2	90%	52%
25	QGA.9	General Aptitude	Quantitative aptitude — Logarithms — change of base	MCQ	1	90%	50%
26	QGA.11	General Aptitude	Quantitative aptitude — Integers satisfying an inequality	MCQ	1	90%	48%

Section 10 — What Not to Skip, and What to Skip

There are two ways to be wrong about a syllabus. You can study something it does not contain, which costs time. You can skip something it does contain, which costs marks. **These are not symmetric**, and treating them as though they were is the mistake behind most reading lists.

The asymmetry has a practical consequence that runs through this whole chapter: where a topic's status is genuinely unclear, read it briefly rather than skip it. An hour lost is recoverable. A question you cannot attempt is not.

This chapter gives two lists of what not to skip, built from opposite kinds of evidence, and then says what can be skipped. They are not mirror images of each other.

10.1 Do not skip, group one: the topics examined in all six counted papers

The strongest signal six papers can give is a topic that drew marks in **every one of them** — 2022 S1, 2022 S2, 2023, 2024, 2025 and 2026, five years, five different setting institutes, no gap. Six papers show patterns, not trends, and this is the clearest pattern they show: a topic the examiners have not once left out.

18 of the 91 topic groups in the counted record did so, 14 of them technical. They carry 238 of the 600 marks counted across the six papers. If your preparation is incomplete, it should be incomplete somewhere other than here.

Mean a year averages the sets of a year first, so no year counts twice.

Topic	Section	Counted marks	Mean a year	Papers
Quantitative aptitude: computation, geometry, statistics, probability	General Aptitude	29	4.4	6 of 6
Verbal aptitude	General Aptitude	20	3.6	6 of 6
Analytical aptitude	General Aptitude	21	3.2	6 of 6
Mechanics of machining, cutting tools, tool life and economics of machining	Materials, Manufacturing and Metrology	18	2.8	6 of 6
Spatial aptitude	General Aptitude	15	2.8	6 of 6
Engineering materials: crystal structure, phase diagrams, heat treatment, stress-strain	Materials, Manufacturing and Metrology	14	2.4	6 of 6
Calculus: limits, continuity, integrals, partial derivatives, Taylor series, maxima and minima, Fourier series	Engineering Mathematics	14	2.3	6 of 6
Second law, entropy, availability and irreversibility	Thermodynamics and Applications	13	2.1	6 of 6
Convection: boundary layer, dimensionless parameters, correlations for plates and pipes, turbulence	Heat Transfer	11	1.8	6 of 6
Radiation: Stefan-Boltzmann and Wien's laws, grey surfaces, view factors, networks	Heat Transfer	10	1.8	6 of 6
Free vibration and damping (single and two degrees of freedom)	Theory of Machines and Vibrations	10	1.8	6 of 6
Complex variables: analytic functions, Cauchy-Riemann, Cauchy's integral theorem, series	Engineering Mathematics	10	1.6	6 of 6
Plane trusses and frames	Engineering Mechanics	10	1.6	6 of 6
Fluid acceleration, continuity equation and velocity potential	Fluid Mechanics	10	1.6	6 of 6
Inventory control	Industrial Engineering	9	1.6	6 of 6

Topic	Section	Counted marks	Mean a year	Papers
Vector calculus: gradient, divergence, curl, line/surface/volume integrals, Gauss, Stokes, Green	Engineering Mathematics	8	1.4	6 of 6
CAD/CAM, NC/CNC and automation	Materials, Manufacturing and Metrology	8	1.4	6 of 6
Machining operations, abrasive and non-traditional machining, work holding	Materials, Manufacturing and Metrology	8	1.4	6 of 6

A caution about the topics *not* in that table

****A low paper-count is much weaker evidence than a high one, and this book draws no skip conclusion from it.****
There are two reasons.

The first is the size of the record. With six papers, a topic set once is as consistent with *rare* as with *not drawn yet*, and nothing in the record can tell the two apart.

The second is mechanical. Every counted question is placed in exactly one section and one topic, and ME's sections share their methods, so a skill used inside questions counted elsewhere shows a low count of its own without being rare in the examination. The sharpest case in the record is Machine Design in 2023: the section was counted at **2 marks**, while 4 more marks in the same paper — 2023 (Q29), 2023 (Q34) and 2023 (Q60) — used it as the working tool and were counted where their method sits. A reader of the section total alone would conclude that the paper hardly asked for it. It asked for it more than the total shows.

Read the table above as *these are certain*. Do not read its absences as *these are safe*.

10.2 Do not skip, group two: what the syllabus names and no paper has set

These have no past-paper history at all, so ****no amount of previous-year practice will make you meet them****. That is exactly why they belong on a do-not-skip list: they are the material most likely to be missing from a preparation built on past papers.

First, the whole topic groups that have drawn no marks in the six papers:

Section	Topic named in the syllabus, with no counted marks
Engineering Mechanics	Virtual work
Mechanics of Materials	Strain gauges and rosettes
Mechanics of Materials	Mechanical properties: toughness, hardness, impact strength
Theory of Machines and Vibrations	Cams
Theory of Machines and Vibrations	Critical speeds of shafts
Theory of Machines and Vibrations	Control systems: PID controller, transfer function
Machine Design	Gears and belt drives
Machine Design	Brakes and clutches
Fluid Mechanics	Fluid properties
Heat Transfer	Boiling and condensation
Thermodynamics and Applications	Combustion: air-fuel ratio, equivalence ratio
Materials, Manufacturing and Metrology	Non-destructive testing
Industrial Engineering	Work study, productivity, quality, reliability, six sigma, lean

Section 11 — The Ten Mock Tests

Ten full papers follow. Each is 65 questions, 100 marks and three hours, built to the GATE Mechanical Engineering pattern and to the section weights of the six counted papers. Every question in them was written for this book; none reproduces a past paper and none repeats a question from the subject chapters, which is why no mock question carries a year as a source.

A mock test is not a question bank with a clock attached. It measures three things the bank cannot: whether you recognise a topic when nothing announces it, whether you can hold accuracy for three hours, and where your time actually goes as opposed to where you think it goes. Use them for that.

11.1 The common blueprint

All ten papers are laid out identically, because the real paper is: all six counted papers — 2022 S1, 2022 S2, 2023, 2024, 2025 and 2026 — share this layout exactly.

Questions	Block	Marks each
Q1 – Q5	General Aptitude	1
Q6 – Q10	General Aptitude	2
Q11 – Q35	Technical	1
Q36 – Q65	Technical	2

That gives 30 one-mark and 35 two-mark questions, General Aptitude at exactly 15 marks and the technical sections at 85 — the shape all six counted papers close on. In all six, every General Aptitude question was a multiple-choice question, 60 of 60, and the mocks keep that too.

The 85 technical marks are distributed as the six counted papers distribute them. The target for each section is its mean a year — the two 2022 sets averaged first, so that 2022 counts once, then the five years averaged — rounded to whole marks so that the ten targets sum to 85. Where two sections are equally near the next mark, the larger section takes it.

Section	2022 S1	2022 S2	2023	2024	2025	2026	Mean a year	Target in each mock
Materials, Manufacturing and Metrology	15	17	21	16	20	14	17.4	18
Engineering Mathematics	13	13	13	13	13	13	13.0	13
Thermodynamics and Applications	8	8	7	8	11	10	8.8	9
Mechanics of Materials	6	6	15	6	7	8	8.4	8
Fluid Mechanics	8	8	10	7	8	6	7.8	8
Heat Transfer	8	8	7	9	5	8	7.4	7
Industrial Engineering	8	8	2	8	6	10	6.8	7
Theory of Machines and Vibrations	8	8	2	8	8	7	6.6	7
Engineering Mechanics	8	6	6	6	2	6	5.4	5
Machine Design	3	3	2	4	5	3	3.4	3
Technical total	85	85	85	85	85	85	85.0	85

Three consequences are worth saying out loud. **The mocks vary the weights a little, but far less than the real paper has.** A mock may move a section by a mark or two from its target, as a real paper does, and never by more than three. The six real papers moved further: Mechanics of Materials ran from 6 marks to 15, Industrial Engineering from 2 to 10, and Theory of Machines and Vibrations and Engineering Mechanics each from 2 to 8. A practice paper that swung that far would mix two measurements at once — your preparation and the paper's shape — so the mocks do not, but it means no mock can show you how a fifteen-mark Mechanics of Materials year feels.

Engineering Mathematics and General Aptitude do not vary at all, in the mocks or in the record:

Engineering Mathematics carried exactly 13 marks in nine questions, five at one mark and four at two, in all six papers, and General Aptitude exactly 15.

And the weights are history, while the syllabus is not. The targets are counted from papers set before the 2027 syllabus added its 30 phrases, so they say nothing about how much of a section's marks the newly named topics will draw. The weights the mocks are held to are the only weights there is evidence for, and they should be read as that and no more.

Marking, exactly as the official keys apply it.

Type	Correct	Wrong	Partly correct
MCQ	full marks	minus one-third of the question's marks	not possible
MSQ	full marks	no penalty	no credit
NAT	full marks for any value in the band	no penalty	not applicable

So a wrong one-mark MCQ costs $1/3$ and a wrong two-mark MCQ costs $2/3$. MSQ scores only when every correct option is selected and no incorrect one is — no partial credit, no negative marking — which makes it the format that punishes partial knowledge hardest. NAT carries no penalty at all, so a NAT left blank is strictly worse than a NAT attempted.

Each paper states its own conventions — SI units throughout, $g = 9.81 \text{ m/s}^2$ and water density 1000 kg/m^3 unless a question says otherwise — and every property or table value a question needs is printed in the question: air constants, steam-table, refrigerant-table and psychrometric values, material properties, the Stefan-Boltzmann constant, any standard normal value. No question depends on a drawing. Mechanisms, free-body diagrams, beams, trusses, pipe networks, cycles and part geometry are described in words — dimensions, positions, directions, supports and loads — so that every question can be solved from its text alone. Every numerical-answer question prints its acceptance band.

11.2 Why the MSQ and NAT counts vary between papers

The number of multiple-select and numerical-answer questions is different in each of the ten papers, and that is deliberate, because it is different in each real paper. Counted from the official answer keys:

Paper	MCQ	MSQ	NAT
2022 S1	35	6	24
2022 S2	35	6	24
2023	32	8	25
2024	39	2	24
2025	38	1	26
2026	45	0	20

Because General Aptitude has been all MCQ, every MSQ and every NAT in those papers sat in the technical 85 marks. The mocks stay inside the counted ranges — 32–45 MCQ, 0–8 MSQ and 20–26 NAT in a paper — and move within them from paper to paper. The multiple-select question fell from 8 in 2023 to 0 in 2026, so most of the ten carry few or none and one or two carry more; each paper says at its head how many of each it holds. A set of mocks with a constant mix would train you for a paper that does not exist. Varying it means some of these ten papers will feel harsher than others for reasons that have nothing to do with your preparation — which is exactly what the record supports.

Mock Test 01

Duration: 180 minutes | Maximum marks: 100 | 65 questions

The layout of the real ME paper: Q1–Q5 General Aptitude at 1 mark, Q6–Q10 General Aptitude at 2 marks, Q11–Q35 technical at 1 mark, Q36–Q65 technical at 2 marks. Inside each technical block the multiple-choice questions come first, then the multiple-select, then the numerical-answer questions, as in the real paper.

This paper carries two multiple-select questions and twenty-four numerical-answer questions, with thirty-nine multiple-choice questions — a mix from the middle of the ranges of the six counted ME papers (MSQ 0 to 8, NAT 20 to 26). Every section sits exactly on its counted weight, so this is the paper to take first: its score tells you where you stand against the average shape of the real paper, before the later mocks begin to move the weights about. Sixteen of the thirty two-mark questions are numerical, and none of them can lose you a mark, so never leave a NAT blank.

Marking. MCQ: correct +marks, wrong $-1/3$ of marks. MSQ: all correct options and no incorrect one, or nothing; no negative marking. NAT: no negative marking; any value in the printed band scores.

Conventions. SI units; $g = 9.81 \text{ m/s}^2$ and water density 1000 kg/m^3 unless a question says otherwise; every property and table value needed is given in the question. Angles are in degrees unless radians are stated. No question depends on a drawing: every beam, mechanism, tank and mould is described in words.

Q1 – Q5 carry ONE mark each.

Q1 MCQ · 1 mark

Select the option that completes the sentence correctly.

"The endurance rig _____ without a stop since Monday morning, and the log shows no fault so far."

- (A) runs
- (B) is running
- (C) has been running
- (D) had been running

General Aptitude — grammar: the present perfect continuous with "since"

Q2 MCQ · 1 mark

Butter : Milk :: Paper : _____

- (A) Book
- (B) Pulp
- (C) Ink
- (D) Printer

General Aptitude — word analogy: a product and the material it is made from

Q3 MCQ · 1 mark

The number of zeros at the end of $60!$, the product of all the integers from 1 to 60, is

- (A) 6
- (B) 12
- (C) 56
- (D) 14

General Aptitude — numbers: zeros at the end of a factorial

Q4 MCQ · 1 mark

Nine ball bearings look identical. Eight of them have the same mass and the ninth is slightly heavier. Only a two-pan balance is available, with no weights. The least number of weighings that is certain to find the heavier bearing, however the weighings turn out, is

- (A) 2 (B) 3
(C) 4 (D) 8

General Aptitude — analytical reasoning: finding the odd one with a balance

Q5 MCQ · 1 mark

A solid block 6 units long, 4 units wide and 3 units high is built from 72 unit cubes and stands on a table. It is viewed from a point from which exactly three of its faces can be seen: the top, the front and one side. The number of unit cubes that show at least one face to the viewer is

- (A) 41 (B) 42
(C) 54 (D) 72

General Aptitude — spatial reasoning: cubes visible from a corner

Q6 – Q10 carry TWO marks each.

Q6 MCQ · 2 marks

The table gives the number of staff in the three departments of a firm in two successive years, and the percentage of the staff of each department who are women.

Department	Staff, year 1	Women, year 1 (%)	Staff, year 2	Women, year 2 (%)
Design	40	25	50	30
Production	120	10	100	15
Testing	40	50	50	40

From year 1 to year 2, the total number of women in the three departments

- (A) does not change (B) rises by about 8 per cent
(C) rises by about 19 per cent (D) rises by about 4 per cent

General Aptitude — data interpretation: percentages taken on changing bases

Q7 MCQ · 2 marks

A motor boat takes 6 hours to travel 24 km upstream and 36 km downstream, and 6.5 hours to travel 36 km upstream and 24 km downstream. The speed of the boat in still water and the speed of the stream are both constant. The speed of the stream is

- (A) 2 km/h (B) 4 km/h
(C) 8 km/h (D) 10 km/h

General Aptitude — time, speed and distance: boats and streams

Q8 MCQ · 2 marks

The four sentences P, Q, R and S below, when put in the right order, make a coherent paragraph.

Q62 NAT · 2 marks

A flat plate casting 600 mm × 450 mm in plan is moulded wholly in the drag, with its upper face in the parting plane. The mould is filled with molten iron of density 7100 kg/m³ up to a level 180 mm above the parting plane. The cope, with its sand, weighs 2.2 kN. Take $g = 9.81 \text{ m/s}^2$. The least additional weight that must be placed on the cope to keep the metal from lifting it is ____ N (round off to the nearest integer).

Acceptance band: 1170 to 1200.

Materials, Manufacturing and Metrology — patterns, moulds and cores: the lifting force on a cope

Q63 NAT · 2 marks

A metal cylinder 60 mm in diameter and 90 mm high is upset in a press, between flat platens, to a height of 54 mm. Friction at the platens is neglected, so the cylinder stays a cylinder as it is squeezed. The flow curve of the metal is $\sigma = 620 \epsilon^{0.22} \text{ MPa}$, σ being the true stress and ϵ the true strain. The work done by the press is ____ kJ (round off to one decimal place).

Acceptance band: 56.0 to 58.0.

Materials, Manufacturing and Metrology — bulk forming: the ideal work of deformation

Q64 NAT · 2 marks

Two plates, each 12 mm thick, are joined by a single-V butt weld 1 m long. The groove has an included angle of 60° and a root gap of 2 mm, with no root face, and the weld is finished flush with the plate surfaces. Each electrode has a core wire 4 mm in diameter and 450 mm long; a stub 40 mm long is thrown away from each, and 15 per cent of the core metal that is melted is lost as spatter and fume. The least number of whole electrodes needed is ____ (answer in integer).

Acceptance band: exactly 25.

Materials, Manufacturing and Metrology — principles of welding: filler metal for a single-V butt joint

Q65 NAT · 2 marks

Lots are sentenced by a single sampling plan: a random sample of 80 items is taken from each lot, and the lot is accepted if the sample contains 2 defective items or fewer. A lot in which 2.5 per cent of the items are defective is submitted. Using the Poisson approximation for the number of defectives in the sample, the probability that the lot is accepted is ____ (round off to three decimal places).

Acceptance band: 0.672 to 0.682.

Industrial Engineering — quality concepts: probability of acceptance under a sampling plan

Answer key — Mock Test 01

Attempt the whole paper before reading this page. Score with the official scheme: a correct 1-mark question gives +1 and a correct 2-mark question +2; a wrong MCQ costs 1/3 of a mark (1-mark) or 2/3 of a mark (2-mark); MSQ and NAT carry no negative marking, and an MSQ scores only if every correct option is selected and no incorrect one is. NAT answers are given to the precision stated in each question.

This paper contains 39 MCQ, 2 MSQ and 24 NAT questions (30 one-mark and 35 two-mark, 100 marks).

Q	Answer	Type	Q	Answer	Type	Q	Answer	Type
1	C	MCQ	23	A	MCQ	45	C	MCQ
2	B	MCQ	24	D	MCQ	46	B	MCQ
3	D	MCQ	25	C	MCQ	47	A	MCQ
4	A	MCQ	26	D	MCQ	48	A	MCQ
5	B	MCQ	27	A,C	MSQ	49	A,C,D	MSQ
6	C	MCQ	28	0.208	NAT	50	0.438	NAT

Section 13 — Detailed Mock Test Solutions

Each test's solutions follow, in question order. Attempt and score the paper (Section 12) before reading these.

Mock Test 01

Q1**ANSWER C**

Confidence: Pattern 51% | Exact-question 11%

"Since Monday morning" names the point at which an action began, and "the log shows no fault so far" tells us the action is still going on. An action that began in the past and continues to the present takes the present perfect continuous: **has been running**.

A. runs is the simple present, which states a habit and cannot carry "since". **B. is running** describes what is happening now but cannot reach back to a starting point, so it does not go with "since" either. **D. had been running** is the past perfect continuous; it needs a later past moment to look back from, and the second clause, "shows", is in the present.

Q2**ANSWER B**

Confidence: Pattern 51% | Exact-question 11%

Butter is made from milk: the first word is a product and the second is the material it is made from. Paper is made from **pulp**.

A. Book is something made from paper, which turns the relation round. **C. Ink** is used on paper and **D. Printer** is a machine that works on paper; neither is what paper is made from.

Q3**ANSWER D**

Confidence: Pattern 51% | Exact-question 11%

Each zero at the end of a number needs one factor of 10, that is one 2 and one 5. In 60! the factors of 2 are plentiful, so the number of zeros is the number of factors of 5.

- Multiples of 5 up to 60: $60/5 = 12$, each giving one factor of 5
- Multiples of 25 up to 60: 25 and 50, each giving one more

Total $12 + 2 = 14$.

B. 12 counts the multiples of 5 and forgets that 25 and 50 each hold a second 5. **A. 6** counts only the multiples of 10. **C. 56** is the number of factors of 2 in 60! ($30 + 15 + 7 + 3 + 1$), which is not the limiting count.

Q4**ANSWER A**

Confidence: Pattern 51% | Exact-question 11%

A weighing has three possible results: the left pan goes down, the right pan goes down, or the pans balance. So each weighing can split the bearings into three groups.

- First weighing: put three bearings on each pan and leave three aside. The heavier bearing is in the pan that goes down, or in the three left aside if the pans balance.
- Second weighing: from that group of three, put one on each pan. The pan that goes down holds the heavier bearing; if they balance it is the third.

2 weighings are enough, and one is not: a single weighing has only three outcomes and cannot pick one bearing out of nine.

- B. 3** comes from halving the bearings each time (four against four, two against two, one against one), which wastes the third outcome. **C. 4** comes from weighing the bearings in pairs, one against one.
- D. 8** compares the bearings one at a time with a chosen reference.

Q5**ANSWER B**

Confidence: Pattern 51% | Exact-question 11%

A unit cube is hidden only if it lies on none of the three visible faces, that is, if it is not in the top layer, not in the front layer and not in the visible side layer. Removing one layer in each direction leaves a hidden block of $(6 - 1) \times (4 - 1) \times (3 - 1) = 5 \times 3 \times 2 = 30$ cubes. The number visible is $72 - 30 = 42$.

Check by inclusion and exclusion: the three faces show $24 + 18 + 12 = 54$ squares; the cubes along the three edges where two visible faces meet ($6 + 4 + 3 = 13$) have been counted twice, and the corner cube where all three meet has then been removed once too often: $54 - 13 + 1 = 42$.

C. 54 adds the three faces with no correction, counting the edge cubes twice. **A. 41** subtracts the edges but does not put the corner cube back. **D. 72** counts every cube in the block, hidden or not.

Q6**ANSWER C**

Confidence: Pattern 51% | Exact-question 11%

Turn each percentage into a number of women.

Department	Women, year 1	Women, year 2
Design	25% of 40 = 10	30% of 50 = 15
Production	10% of 120 = 12	15% of 100 = 15
Testing	50% of 40 = 20	40% of 50 = 20
Total	42	50

The rise is 8 on 42, that is $8/42 = 0.190$: the number **rises by about 19 per cent**.

(A) is the trap in the table: the total staff is 200 in both years, and the plain average of the three percentages is 28.3 in both years, yet the number of women changes because the percentages apply to different bases. **(B)** gives the rise in the number of women, 8, as if it were a percentage. **(D)** is the change in the share of women in the whole staff, from $42/200 = 21$ per cent to $50/200 = 25$ per cent; it is a difference of 4 percentage points, not a rise of 4 per cent.

Q7**ANSWER A**

Confidence: Pattern 51% | Exact-question 11%

Let the speeds upstream and downstream be u and d km/h.

- $24/u + 36/d = 6$
- $36/u + 24/d = 6.5$

Adding: $60(1/u + 1/d) = 12.5$, so $1/u + 1/d = 0.20833$. Subtracting the first from the second:

$12(1/u - 1/d) = 0.5$, so $1/u - 1/d = 0.04167$. Hence $1/u = 0.125$ and $1/d = 0.08333$, giving $u = 8$ km/h and $d = 12$ km/h.

Speed of the stream = $(d - u)/2 = (12 - 8)/2 = 2$ km/h; the boat does 10 km/h in still water. Check:

$24/8 + 36/12 = 3 + 3 = 6$ h, and $36/8 + 24/12 = 4.5 + 2 = 6.5$ h.

B. 4 km/h is $d - u$, not halved. **C. 8 km/h** is the speed upstream. **D. 10 km/h** is the speed of the boat in still water.

Section 15 — The Last-Minute Sheet

What follows is the set of formulae, constants, rules of thumb and facts worth carrying into the last day, grouped by the ten technical sections of this book. Each group opens with the 2027 syllabus's own text for the paragraphs it covers, quoted as printed, so that the sheet can be checked against what the paper may ask: every topic the syllabus names has an entry here, including the ones no counted paper has examined and the ones named for the first time in 2027. Every numerical example on the sheet was recomputed before printing. General Aptitude has its own syllabus and its own chapter, and is not repeated here.

Two things this sheet deliberately is not. **It is not a syllabus course.** A formula list cannot tell you what a formula is for, and a candidate meeting one of these for the first time on this sheet is meeting it too late. **And it does not replace the chapters.** Each entry is the short form of something the subject chapter works through in full; where a result has a condition — a thin wall, a small damping ratio, an isentropic process, a constant failure rate — the condition is printed beside it, because a result used outside its condition gives a confident wrong answer. Where a value is a rule of thumb rather than a law, it says so; the question's own data always wins.

Notation: SI units throughout; $g = 9.81 \text{ m/s}^2$. $\ln$ is the natural logarithm. Angles inside formulae are in radians unless degrees are shown. Temperatures inside any ratio, power or exponential are absolute, in kelvin. A dot over a symbol is a rate ($\dot{m}$, $\dot{Q}$, $\dot{W}$). Subscript 1 is an inlet or an initial state and 2 an exit or a final state unless stated.

15.1 Engineering Mathematics

Linear Algebra

Syllabus — Linear Algebra. Matrix algebra, systems of linear equations, eigen values and eigen vectors.

Item	Result
Systems $Ax = b$, n unknowns	unique solution iff $\text{rank } A = \text{rank } [A : b] = n$; infinitely many iff $\text{rank } A = \text{rank } [A : b] < n$; none iff $\text{rank } A < \text{rank } [A : b]$
Homogeneous $Ax = 0$	always consistent; a non-trivial solution iff $\text{rank } A < n$ (square A : iff $\det A = 0$)
Rank	number of non-zero rows in echelon form; $\text{rank } A = \text{rank } A^T$; $\text{rank}(AB) \leq \min(\text{rank } A, \text{rank } B)$
Determinant	$\det(AB) = \det A \cdot \det B$; $\det A^T = \det A$; $\det(kA) = k^n \det A$ for $n \times n$; triangular: product of the diagonal; a row swap changes the sign; two equal rows give 0
Inverse	$(AB)^{-1} = B^{-1}A^{-1}$; $A^{-1} = \text{adj } A / \det A$; 2×2 : $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = (1/(ad - bc)) \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$; $\det(\text{adj } A) = (\det A)^{n-1}$
Eigenvalues	$\det(A - \lambda I) = 0$; $\sum \lambda = \text{trace}$, $\prod \lambda = \det$; 2×2 : $\lambda^2 - (\text{trace})\lambda + \det = 0$
Spectral mapping	A^k has λ^k , A^{-1} has $1/\lambda$, $A + cI$ has $\lambda + c$, kA has $k\lambda$, same eigenvectors
Special matrices	triangular: eigenvalues on the diagonal; real symmetric: real eigenvalues, orthogonal eigenvectors; skew-symmetric: zero or purely imaginary eigenvalues; orthogonal ($A^T A = I$): every eigenvalue of modulus 1, $\det = \pm 1$
Cayley–Hamilton	every square matrix satisfies its own characteristic equation — used to find A^{-1} or A^n
Independence	n vectors in $\mathbb{R}^n$ are independent iff their matrix has a non-zero determinant; eigenvectors of distinct eigenvalues are independent
Example	$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$: eigenvalues 1 and 3, eigenvectors along $(1, -1)$ and $(1, 1)$

Calculus

Syllabus — Calculus. Functions of single variable, limit, continuity and differentiability, mean value theorems, indeterminate forms; evaluation of definite and improper integrals; double and triple integrals; partial derivatives, total derivative, Taylor series (in one and two variables), maxima and minima, Fourier series; gradient, divergence and curl, vector identities, directional derivatives, line, surface and volume integrals, applications of Gauss, Stokes and Green's theorems.

Item	Result
Standard limits, $x \rightarrow 0$	$\sin x/x \rightarrow 1$; $(1 - \cos x)/x^2 \rightarrow 1/2$; $(e^x - 1)/x \rightarrow 1$; $\ln(1 + x)/x \rightarrow 1$; $(1 + x)^{1/x} \rightarrow e$
Indeterminate forms	$0/0$ and ∞/∞ : L'Hôpital, $\lim f/g = \lim f'/g'$; $0 \cdot \infty$ and $\infty - \infty$: rewrite as a quotient; 1^∞ , 0^0 , ∞^0 : take logs — $\lim f^g = e^{(\lim g \ln f)}$, and for 1^∞ , $e^{(\lim g(f - 1))}$
Continuity, differentiability	continuous at a when the left limit, the right limit and $f(a)$ agree; differentiable $\Rightarrow$ continuous, never the reverse (the modulus of x at 0)
Mean value theorems	Rolle: $f(a) = f(b) \Rightarrow f'(c) = 0$ for some c in (a, b) ; Lagrange: $f'(c) = (f(b) - f(a))/(b - a)$; Cauchy: $f'(c)/g'(c) = (f(b) - f(a))/(g(b) - g(a))$
Definite integrals	$\int_0^a f(x) dx = \int_0^a f(a - x) dx$; over $(-a, a)$: 0 for odd f , $2 \int_0^a f$ for even f
Wallis	$\int_0^\pi (\pi/2) \sin^n x dx = \int_0^\pi (\pi/2) \cos^n x dx = [(n - 1)(n - 3)...]/[n(n - 2)...]$, times $\pi/2$ when n is even
Gamma function	$\Gamma(n + 1) = n\Gamma(n) = n!$; $\Gamma(1/2) = \sqrt{\pi}$; $\int_0^\infty x^n e^{-ax} dx = n!/a^{n+1}$; $\int_0^\infty e^{-x^2} dx = \sqrt{\pi}/2$
Improper integrals	$\int_1^\infty dx/x^p$ converges iff $p > 1$; $\int_0^1 dx/x^p$ converges iff $p < 1$
Double and triple integrals	change the order by redrawing the region; polar $dx dy = r dr d\theta$; cylindrical $r dr d\theta dz$; spherical $r^2 \sin \phi dr d\phi d\theta$; in general multiply by the Jacobian
Partial derivatives	Euler's theorem for f homogeneous of degree n : $x f_x + y f_y = n f$; total derivative $df/dt = f_x dx/dt + f_y dy/dt$
Taylor series, one variable	$f(a + h) = f(a) + h f'(a) + (h^2/2!) f''(a) + \dots$; $e^x = \sum x^n/n!$; $\sin x = x - x^3/3! + \dots$; $\cos x = 1 - x^2/2! + \dots$; $\ln(1 + x) = x - x^2/2 + x^3/3 - \dots$
Taylor series, two variables	$f(a + h, b + k) = f + h f_x + k f_y + \frac{1}{2}(h^2 f_{xx} + 2hk f_{xy} + k^2 f_{yy}) + \dots$, derivatives at (a, b)
Maxima and minima, one variable	$f' = 0$; $f'' > 0$ minimum, $f'' < 0$ maximum; on a closed interval also compare the end values
Maxima and minima, two variables	$f_x = f_y = 0$; with $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$: $rt - s^2 > 0$ and $r > 0$ minimum, $r < 0$ maximum; $rt - s^2 < 0$ saddle point; $rt - s^2 = 0$ no conclusion
Constrained extremum	Lagrange multiplier: $\nabla f = \lambda \nabla g$ at the optimum of f subject to $g = 0$
Fourier series, period $2L$	$f(x) = a_0/2 + \sum [a_n \cos(n\pi x/L) + b_n \sin(n\pi x/L)]$, $a_n = (1/L) \int_{-L}^L f \cos(n\pi x/L) dx$, $b_n = (1/L) \int_{-L}^L f \sin(n\pi x/L) dx$; even f : cosines only; odd f : sines only; at a jump the series converges to the mean of the two one-sided limits
Gradient, directional derivative	$\nabla \phi$ is normal to the level surface $\phi = c$; rate of change along unit vector $\hat{a}$ is $\nabla \phi \cdot \hat{a}$; greatest rate is the magnitude of $\nabla \phi$, along $\nabla \phi$
Divergence and curl	$\nabla \cdot F = \partial F_1/\partial x + \partial F_2/\partial y + \partial F_3/\partial z$; solenoidal: $\nabla \cdot F = 0$; irrotational: $\nabla \times F = 0$, then $F = \nabla \phi$ and line integrals are path-independent
Vector identities	$\nabla \times (\nabla \phi) = 0$; $\nabla \cdot (\nabla \times F) = 0$; $\nabla \cdot (\phi F) = \phi \nabla \cdot F + F \cdot \nabla \phi$; $\nabla \times (\nabla \times F) = \nabla(\nabla \cdot F) - \nabla^2 F$; $\nabla \cdot r = 3$ and $\nabla \times r = 0$ for $r = (x, y, z)$
Gauss (divergence)	$\oint F \cdot n dS$ over a closed surface = $\int \nabla \cdot F dV$ over the enclosed volume
Stokes	$\oint F \cdot dr$ round a closed curve = $\int (\nabla \times F) \cdot n dS$ over any surface it bounds
Green	$\oint (M dx + N dy) = \iint (\partial N/\partial x - \partial M/\partial y) dx dy$, anticlockwise; area enclosed = $\frac{1}{2} \oint (x dy - y dx)$
Example	flux of $F = (x, y, z)$ out of a sphere of radius R : $\nabla \cdot F = 3$, so $3 \times (4/3)\pi R^3 = 4\pi R^3$

Section 16 — Sources, Conventions and Limits

This chapter says what the book was built from, how the counting was done, and what the book does not claim. It is here so that a reader who doubts a figure knows which file to open.

16.1 What the book was built from

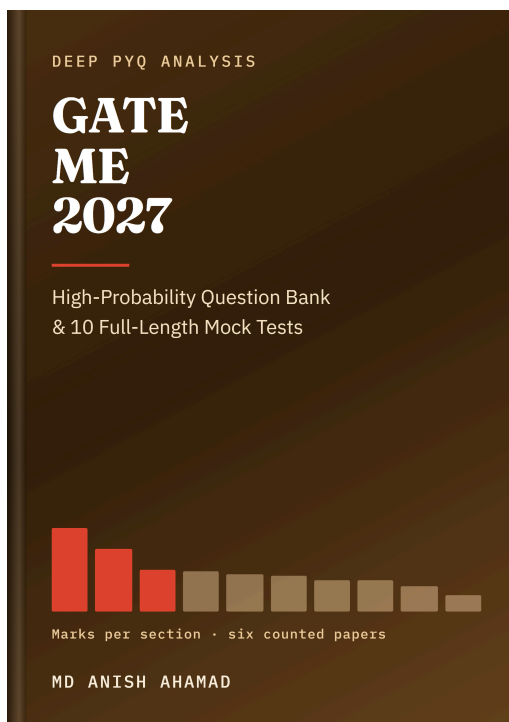
Four official syllabus documents. The GATE Mechanical Engineering syllabi of 2024 (IISc Bengaluru), 2025 (IIT Roorkee), 2026 (IIT Guwahati) and 2027 (IIT Madras), each held as the organiser's own PDF in `research/official/`. Section 2 reproduces the 2027 document and compares it with the 2026 one phrase by phrase; the comparison is produced by a program that reads the PDFs, not from memory, and it finds 30 phrases added, 5 removed and 19 groups reworded, with two paragraph labels new and two gone. The same program finds the 2024, 2025 and 2026 documents identical in wording, differing only in the capitalisation or hyphen of three paragraph labels between 2024 and 2025. The 2022 and 2023 syllabus documents were not obtained, and this book claims nothing about them.

Six official question papers and their official answer keys — every GATE ME paper of the five years 2022 to 2026. ME was set in two sets in 2022, both organised by IIT Kharagpur, and as a single paper in each later year: 2023 (IIT Kanpur), 2024 (IISc Bengaluru), 2025 (IIT Roorkee) and 2026 (IIT Guwahati). That is 390 questions and 600 marks. A citation in this book names the paper and its printed question number: "2025 (Q35)" for a year with one paper, and "2022 S1 (Q35)" or "2022 S2 (Q35)" for 2022, whose set is always named. Every file was downloaded from a GATE organising-institute website. No coaching site, aggregator, question repository or blog was used for any paper, key, syllabus or figure in this book.

File	Source
2022 question papers, both sets (one file)	https://gate2026.iitg.ac.in/doc/download/2022/me_2022.pdf
2022 answer keys, both sets (one file)	https://gate2026.iitg.ac.in/doc/download/Answer_keys2022/me-merged_2022.pdf
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2027 General Aptitude syllabus	https://gate2027.iitm.ac.in/static/doc/GATE2027_Syllabus/GA_GATE2027_Syllabus.pdf

All were downloaded on 30 September 2026. Four things about them a reader should have rather than discover:

- **The 2022, 2023 and 2024 papers and keys come from the IIT Guwahati (GATE 2026) site's archive.** The 2022 and 2023 organisers' own GATE sites were tried on the same day and could not be reached. The 2024 file's name ends "S2" because ME was sat in session 2 that year; it is the only 2024 ME paper, not a second set.



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